Nightingale Care Note

Technical Brief

LIVE DEMO nightingale-care-note.vercel.app

1. Why I built it this way

During a consultation, a clinician may have months of notes to review. Lab results sit beside staff follow ups, patient messages, and earlier consult summaries. The useful detail is present, but finding what matters for today's visit takes time.

Nightingale Care Note gives the clinician a shorter starting point. It selects three current priorities, then keeps the supporting record close enough to inspect. The aim is to reduce reading time without asking the clinician to accept an AI summary on trust.

Ranking is deterministic. Important claims retain exact source references. A clinician can confirm or reject ordinary suggestions. Earlier information stays in the record when a later entry corrects it.

2. The 10 second Glance

Sarah Tan is the main longitudinal example. Her clinician view shows dizziness after a medication change, worsening HbA1c, and a pending renal function follow up. Each card includes its time and source. The clinician can open the ranking reason or jump to the evidence.

Risk represents safety severity. A Critical item, or an item with risk at least 0.9, receives a deterministic floor. It stays visible and cannot be dismissed. Importance represents how much workflow attention an item needs now.

```
base = 4*risk + 3*unresolved + 2*recency + 3*clinical_change
+ 2.5*conflict + 1*confirmation

workflow_importance = base * clinic_multiplier
final = max(deterministic_safety_floor, workflow_importance)
```

Pin contributes +3, Accept +2, Source Open +1, Dismiss -2, and Acknowledge 0. The persistent update is:

```
raw_delta = signal * 0.100 / max(4, prior_signal_count + 1)
applied_delta = 0, or sign(raw_delta) * max(abs(raw_delta), 0.001)
multiplier = clamp(previous + applied_delta, 0.80, 1.35)
```

Repeated signals have less effect as exposure grows. The 0.001 floor preserves a nonzero database update. Raw and applied deltas are audited.

Learning changes ordinary workflow order only. It cannot change clinical truth or bypass the deterministic safety floor.

3. What the clinician can check

- Exactly three priorities are shown in Sarah's clinician view.
- View evidence opens the source entry and marks the exact phrase.
- Workflow states are shown instead of an uncalibrated confidence percentage.
- Earlier information remains visible when a clinician later corrects it.

Evidence and patient safety

3. Evidence, grounding, and abstention

Each highlight points to a care entry and one immutable version. Provenance also stores character offsets, the excerpt, and a SHA-256 hash. Before returning a highlight, the API checks the version, offsets, exact text, and hash. A failed check withholds the claim from the normal result.

Raw source -> Structured assertion -> Critical token check -> Display or abstain

Grounding checks medication identity and status. It also checks numbers, dose, units, frequency, laboratory values, and polarity. Allergy assertions include the allergen and reaction. An invalid span or changed critical token causes abstention. Limited semantic support is reported as insufficient evidence to surface confidently.

The interface uses AI Suggested, Clinician Confirmed, Clinician Rejected, Conflict Detected, Needs Review, and Superseded. These states reflect review activity that can be checked. The current provider has no calibrated probability for a confidence percentage.

4. Conflict handling and patient release

Conflict detection covers allergy polarity, medication status, and dosage or frequency. It works across AI and human entries, or across two human entries. The implementation does not claim general medical contradiction detection.

Both assertions and sources remain available after a conflict. The draft enters review, and patient release stays blocked until a clinician resolves the issue. Sarah's metformin example preserves the earlier AI statement while treating the later clinician clarification as authoritative.

Release state is internal, review_required, approved, or released. Patient access requires ownership, patient visibility, released state, a non-AI entry type, and a safe trust state.

> The application query and PostgreSQL Row-Level Security apply the same release boundary.

Live tests allow released patient safe content. They deny another patient, internal records, raw AI notes, internal comments, and internal revision history.

5. Runtime AI path and redaction

POST /api/scribe authenticates the Supabase session and checks clinic scope. Redaction happens before provider processing. Schema validation and critical token grounding follow. Typed conflict checks and the risk floor run before results are separated into grounded, Needs Review, or withheld assertions. The endpoint stores an internal draft with provenance and metadata only audit records.

The provider is deterministic-clinical-v2. No external LLM is called. Using a deterministic provider kept the safety tests and demo results repeatable. A future provider would still pass through the same boundaries.

Redaction fixtures cover known and titled names, including CJK names. They cover Singapore style IDs, phone, email, DOB, multiline address, JSON, and key value text. Tests confirm that medication, dose, frequency, HbA1c, laboratory values, and nonidentifying dates remain available.

A missed identifier is a privacy failure. Removing real clinical detail is an accuracy failure, so the evaluation checks both directions.

Architecture, validation, and limits

6. Architecture and stored data

Browser	->	Next.js on Vercel	->	Auth and domain checks	->	PostgreSQL with RLS

Supabase Auth supplies the identity used by server routes and database policies. The role selector signs in as a seeded synthetic user, so its browser label cannot grant access. PostgreSQL RLS applies clinic scope, role rules, patient ownership, author restrictions, and patient release rules.

The relational model includes clinics, profiles, patients, care entries, immutable versions, comments, tasks, highlights, provenance spans, AI note metadata, feedback, clinic weights, redaction events, and audit logs.

A successful edit increments the entry version. A stale write to the same section returns 409 Conflict. Reverting creates another version that points to the restored one. Audit logs contain actor, action, entity, and version metadata. Note bodies are excluded.

7. Collaboration, Patient View, and Voice Capture

Clinicians and staff can add comments, use mentions, and resolve or reopen threads. Task status persists. Treatment plan edits include diffs and revision history. Undo uses compensating writes. The task API accepts owner changes, but the current interface exposes status only. Reply creates a mention filled follow up instead of a nested reply tree.

Patient View uses simpler language and a smaller information set. Internal comments, raw AI notes, provenance controls, ranking details, and revision history are absent. The family profile flow is a synthetic session only consent prototype.

Voice Capture demonstrates ready, recording, paused, processing, and review states with patient specific fixtures. A clinician can send synthetic text through /api/scribe, which creates an internal draft for hidden QA-0001.

> There is no microphone access, audio upload, speech to text, diarization, live LLM transcription, or production ambient listening.

8. Performance, validation, and limits

The benchmark signs in through /api/session and reuses the returned cookie. It performs 15 warmups and then measures 100 /api/highlights calls. Any non-200 response fails the run.

P50	P95	P99	Failures
207.60 ms	268.50 ms	322.75 ms	0

The production measurement was taken on 28 Aug 2026. P95 is below the required 300 ms target.

Validation reported 0 npm audit vulnerabilities. Lint and typecheck passed. The focused safety suite passed 23 of 23 tests. The live Supabase suite passed 17 of 17 tests for RLS and persistence. A production webpack build passed. Turbopack could not bind its helper port inside the audit sandbox.

The build has no live LLM, real audio processing, generalized clinical NLP, CRDT editor, or EHR integration. It uses synthetic data only. Arbitrary selected text comments were not built. The seed has AI doctor and AI patient session entries, but no AI nurse consult summary.